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(ii) The enthalpy change of the reaction ΔH_r has to be specified with appropriate sign and unit.

(iii) When the chemical reaction is reversed, the value of ΔH is reversed in sign with the same magnitude.

(iv) The physical states (gas, liquid, aqueous, solid in brackets) of all species are important and must be specified in a thermochemical reaction, since ΔH depends on the physical state of reactants and products.

(v) If the thermochemical equation is multiplied throughout by a number, the enthalpy change is also multiplied by the same number.

(vi) The negative sign of ΔH_r indicates that the reaction is exothermic and the positive sign of ΔH_r indicates an endothermic reaction.

For example, consider the following reaction,

$$2\ H_2(g)+O_2(g)\rightarrow 2\ H_2O(g)$$

$$\Delta H_r^0 = -\ 967.4\ kJ$$

$$2\ H_2O\ (g) \rightarrow 2\ H_2(g) + O_2(g)$$

$$\Delta H_r^0 = +\ 967.4\ kJ$$

Standard enthalpy of reaction (ΔH_r^0) from standard enthalpy of formation (ΔH_f^0)

The standard enthalpy of a reaction is the enthalpy change for a reaction when all the reactants and products are present in their standard states. Standard conditions are denoted by adding the superscript 0 to the symbol (ΔH^0)

We can calculate the enthalpy of a reaction under standard conditions from the values of standard enthalpies of formation of various reactants and products. The standard enthalpy of reaction is equal to the difference between standard enthalpy of formation of products and the standard enthalpies of formation of reactants.

$$\Delta H_r^0 = \Sigma\ \Delta H_r^0\ _{(products)} - \Sigma\ \Delta H_r^0\ _{(reactants)}$$

For a general reaction

$$aA + bB \rightarrow cC + dD$$

$$\Delta H_r^0 = \Sigma\ \Delta H_r^0\ _{(products)} - \Sigma\ \Delta H_r^0\ _{(reactants)}$$

$$\Delta H_r^0 = \{c\ \Delta H_f^0\ (C) + d\ \Delta H_f^0\ (D)\} - \{a\ \Delta H_f^0(A) + b\ \Delta H_f^0(B)\}$$

Problem : 7.2

The standard enthalpies of formation of $C_2H_5OH(l)$, $CO_2(g)$ and $H_2O(l)$ are - 277, -393.5 and -285.5 kJ mol^{-1} respectively. Calculate the standard enthalpy change for the reaction

$$C_2H_5OH(l)+3O_2(g)\rightarrow 2CO_2(g)+ 3\ H_2O(l)$$

The enthalpy of formation of $O_2(g)$ in the standard state is Zero, by definition

Solution:

For example, the standard enthalpy change for the combustion of ethanol can be calculated from the standard enthalpies of formation of $C_2H_5OH(l)$, $CO_2(g)$